



Psychology Notes

Scope • Methods • Perception • Psychophysics — Exam Revision

Ch. 1

Scope of Psychology

► What is Psychology?

- **Psychology** = scientific study of behaviour & mental processes
- Goal → describe → explain → predict → influence how people think, feel, act
- Not just clinics/labs — applies to *almost every field* of human life

► Two Broad Divisions (Morgan & King)

Theoretical (Pure) Psychology

- Builds **core knowledge** through research
- Understanding behaviour *for its own sake*
- e.g. Experimental, Physiological, Personality, Abnormal, Social, Developmental Psych.

Applied Psychology

- Puts principles to work on **real problems**
- Serves people directly (clinics, schools, courts, workplace)
- e.g. Clinical, Counseling, Educational, Industrial, Health, Forensic

► Branches — Theoretical

Branch	Focus
Experimental	Lab study of learning, memory, perception, motivation
Physiological	Brain, nervous system, hormones behind behaviour
Developmental	Growth & change across the whole lifespan
Social	Thought/feeling/behaviour in presence of others
Personality	Individual differences — traits, temperament
Abnormal	Types, symptoms, causes of psychopathology

► Branches — Applied

Branch	Focus
Clinical	Assessment, diagnosis, treatment of disorders
Counseling	Everyday personal/educational/career concerns
Educational	Teaching, learning, classrooms
Industrial/Org.	Employer-employee relations, selection, leadership
Health	Psychological factors in illness & wellness
Environmental	Physical surroundings → behaviour/wellbeing
Forensic/Legal	Law, criminal behaviour, justice system
Sports	Mental/emotional factors in athletic performance
Military	Selection, training, morale of armed forces
Consumer	Buying behaviour, decision-making
Community	Wellbeing of communities/social systems

 **Sum up:** Psychology's scope runs from the *neuron* to the *classroom, clinic, courtroom & workplace* — one of the most far-reaching behavioural sciences! (Morgan & King)

Ch. 2 Methods of Psychology

How psychologists actually **study** mind & behaviour → 6 core methods:

Experimental →

Correlational →

Case Study →

Naturalistic Obs. →

Survey/Interview →

Longitudinal/Cross-sectional

▶ 1 Experimental Method

Manipulate **Independent Variable (IV)**, keep rest constant, measure effect on **Dependent Variable (DV)**. **Random assignment** → allows cause-effect claims.

- + Only method to prove causation; high control; replicable
- - Artificial (low real-world validity); ethical limits; reactivity (observer effect)

• **Classic study:** Milgram's Obedience (1963) — ~65% gave max 450V shock under authority pressure.

▶ 2 Correlational Method

Measures 2+ variables as they *naturally occur*; computes correlation coefficient r . Nothing manipulated → **correlation ≠ causation**.

- + Studies variables that can't be manipulated; real-world data; predicts trends
- - Can't prove causation; third-variable problem; direction unclear

• **Classic study:** Doll & Hill — smoking & lung cancer link (1950s), found via correlation not experiment.

▶ 3 Case Study Method

In-depth study of **one** person/group/event — interviews, records, observation.

- + Rich detail; studies rare phenomena; generates hypotheses
- - Doesn't generalize; researcher bias; no cause-effect

• **Classic study:** Phineas Gage (1848) — rod through frontal lobe → personality change → linked brain regions to personality.

▶ 4 Naturalistic Observation

Watch & record behaviour in its **real/natural setting** — no interference.

- + High ecological validity; good for exploratory research

- - No control over variables; observer presence may alter behaviour; no cause-effect

• **Classic study:** Jane Goodall's chimpanzees (Gombe, 1960s) — documented tool use, social bonds.

► 5 Survey & Interview Method

Self-reported data from a sample → generalize to a population.

- + Large samples fast; inexpensive; good for attitudes/beliefs
- - Social desirability bias; depends on honest recall; low response rate skews sample

► 6 Longitudinal vs Cross-Sectional

	<i>Longitudinal</i>	<i>Cross-Sectional</i>
Design	Same participants, over years	Different age groups, one time point
+	Tracks real change, no cohort effect	Fast, cheap, no attrition
-	Costly, slow, attrition	Cohort effects mistaken for age effects

• **Classic study:** Harvard Study of Adult Dev. (1938-present) — warm relationships predict health/happiness more than wealth. Also: Piaget's cognitive-dev theory (from observing his own children).

► Ethics in Research

- Informed Consent → Right to Withdraw → Debriefing → Confidentiality → Protection from Harm

• **Cautionary case:** Stanford Prison Experiment (Zimbardo, 1971) — guards turned abusive, study stopped after 6 days → led to modern ethics boards.

Methods at a Glance

Method	Key Strength	Key Limitation
Experimental	Cause & effect	Artificial, ethical limits
Correlational	Studies unmanipulable vars	No causation
Case Study	Rich detail, rare cases	No generalization
Naturalistic Obs.	Genuine real-world behaviour	No variable control
Survey/Interview	Large, efficient samples	Self-report bias
Longitudinal/Cross-sec.	Change over time / fast	Costly-slow / cohort effects

 Only **experiments** give cause-and-effect. Best method depends on the question: depth (case study) vs breadth (survey) vs realism (naturalistic).

Ch. 3 Perception in Psychology

► Sensation vs Perception

"Sensation is data; Perception is meaning."

Sensation

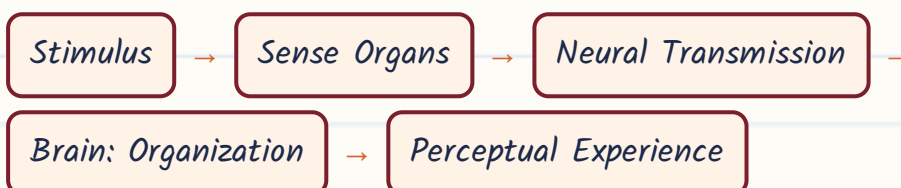
Raw detection of stimuli by sense organs
(eyes, ears, skin) → sent to brain

Perception

Brain **selects, organizes, interprets**
sensory data → meaningful experience

Key qualities of perception: **Selective** · **Subjective** · **Constructive** · **Meaningful**

► Process of Perception



 **Attention** = gatekeeper — only some stimuli get full processing, rest filtered out.

► Attention — What Grabs It?

Stimulus (External) Factors

- Intensity & size
- Contrast
- Movement
- Novelty & repetition

Personal (Internal) Factors

- Motivation & needs
- Interests & values
- Past experience
- Expectation (perceptual set)

► Bottom-Up vs Top-Down Processing

	<i>Bottom-Up</i>	<i>Top-Down</i>
Type	Data-driven	Concept-driven
Key name	J. J. Gibson	Richard Gregory
Idea	Small features → combined into whole percept (part → whole)	Prior knowledge/context → hypothesis → checked against data (whole guides part)

► Five Domains of Perception

1. Perceptual Organization (Gestalt laws, figure-ground)
2. Depth Perception (monocular + binocular cues)
3. Perceptual Constancy (size, shape, colour, brightness)
4. Perceptual Illusions (geometrical, physiological, cognitive)
5. Influencing Factors (personal & stimulus)

► Gestalt Psychology

"The whole is different from the sum of its parts."

Founders: **Wertheimer**, **Köhler**, **Koffka** — mind organizes bits into unified wholes (*Gestalten*).

Law	Meaning
Proximity	Close elements seen as one group
Similarity	Alike elements (colour/shape) grouped together
Closure	Mind fills gaps → sees complete shape
Continuity	Elements in a line/curve seen as continuing smoothly
Figure-Ground	Scene split into focused "figure" + background "ground"

Mnemonic: **P-S-C-C-F** — "Please Sort Circles Carefully, Friend"

► Depth Perception — Monocular Cues (1 eye)

- **Linear Perspective** — parallel lines converge with distance
- **Relative Size** — smaller-looking object = farther
- **Interposition** (overlap) — blocking object = nearer
- **Texture Gradient** — texture denser/finer with distance
- **Aerial Perspective** — distant objects hazier/bluer
- **Light & Shadow** — shading shows form/depth

► Depth Perception — Binocular Cues (2 eyes)

- **Retinal Disparity** — 2 slightly different retinal images → basis of stereopsis
- **Convergence** — inward eye rotation angle signals nearness

► Perceptual Constancy

We perceive objects as **stable** despite retinal image constantly changing.

Type	Example
Size Constancy	Person walking away still "looks" same height
Shape Constancy	Door looks rectangular even when ajar (trapezoid image)
Colour Constancy	Tomato looks red in warm/cool/daylight
Brightness Constancy	White paper looks white in sun & shade; coal looks black in both

► Perceptual Illusions

Illusion = systematic misperception of a real stimulus (vs **Hallucination** = perception with NO stimulus at all).

- **Geometrical** – misjudged length/size/direction (e.g. Müller-Lyer, Ponzo)
- **Physiological** – over-stimulation (e.g. afterimages)
- **Cognitive** – unconscious inference/learned assumptions

• **Müller-Lyer**: arrowheads make equal lines look unequal (size-constancy misapplied).
• **Ponzo**: converging lines make equal bars look unequal (linear-perspective misapplied).
• **Moon Illusion**: moon looks bigger near horizon due to extra terrain distance cues.
• **Ames Room**: distorted room + peephole → equal-height people look very different sizes.
• **Autokinetic effect**: a still light in total darkness appears to drift.

► Factors Influencing Perception

Personal (Subjective)

- Motivation & needs
- Past experience/learning
- Perceptual set
- Emotion/mood
- Culture & upbringing
- Interest & values

Stimulus (Objective)

- Intensity & size
- Contrast
- Movement
- Repetition
- Novelty & familiarity
- Clarity

 Perception = active, selective, constructive. Bottom-up + top-down work together; Gestalt organizes; monocular+binocular give depth; constancies stabilize world; illusions expose the mechanism.

Ch. 4 Psychophysics

► What is Psychophysics?

Psychophysics = quantitative study of relationship between **physical stimuli** (light, sound, pressure) and the **psychological sensations/perceptions** they produce.

Stimulus (Physical)



Sensation (Psychological)

- Founded by **Gustav Fechner** (1860), *Elemente der Psychophysik*
- Considered the **first truly quantitative, experimental branch** of psychology
- Key Q: "How much physical change is needed before a person notices a change?"

► Historical Roots (Timeline)

1834

Ernst Weber
studies JND



1860

Gustav Fechner
Fechner's Law



1950s

Signal Detection Theory

► Absolute Threshold

Absolute Threshold = minimum stimulus intensity detected **50% of the time** — not a fixed on/off point (detection is probabilistic).

Sense	Classic Example (Galanter, 1962)
Vision	Candle flame seen from ~48 km on a clear dark night
Hearing	Watch ticking from ~6 m in silence
Taste	1 tsp sugar in ~7.5 litres of water
Smell	1 drop perfume diffused in a 3-room apartment
Touch	Bee's wing falling 1 cm onto the cheek

► Difference Threshold (JND)

Just Noticeable Difference (JND) = smallest detectable difference between 2 stimuli, 50% of the time.

- 1 kg reference → need +~30g to feel heavier
- 10 kg reference → need +~300g to feel heavier
- **Same 30g is easily felt on 1kg but imperceptible on 10kg** → proportion matters, not absolute amount!

► Weber's Law

$$\Delta I / I = k$$

ΔI = JND (change needed) | I = original intensity | k = Weber fraction (constant)

JND is a **constant proportion** of the original stimulus, not a fixed amount. Smaller k = greater sensitivity.

Sense	Weber Fraction (k)
Vision (brightness)	$\sim 1/60$ (most sensitive)
Weight (heaviness)	$\sim 1/50$
Hearing (loudness)	$\sim 1/10$
Taste (saltiness)	$\sim 1/5$
Smell (odour)	$\sim 1/4$ (least sensitive)

► Fechner's Law

$$S = k \log I$$

S = sensation magnitude | I = stimulus intensity | k = sense-dependent constant

- Built on Weber's Law — integrates ratio rule across many JND steps
- Sensation grows **logarithmically** — equal stimulus *ratios* (e.g. doubling) → equal sensation *increments*

► Signal Detection Theory (SDT)

Classical theory assumes fixed cutoff; SDT says detection also depends on the observer's **decision criterion** (motivation, expectation, cost of errors).

	Signal Present	Signal Absent
Respond "Yes"	✓ Hit	✗ False Alarm
Respond "No"	✗ Miss	✓ Correct Rejection

d' (d-prime) = sensitivity (signal vs noise) | β (beta) = response bias (tendency to say "yes")

► Classical Psychophysical Methods

Method	How it works	Trade-off
Method of Limits	Stimuli presented in ascending/descending steps till response changes	Fast but prone to habituation/expectation errors
Method of Constant Stimuli	Fixed set of stimuli, random order, repeated	Most accurate but time-consuming
Method of Adjustment	Participant adjusts stimulus until just detectable	Quick, intuitive but less precise

► Applications of Psychophysics

- Clinical & Sensory Testing — audiometry, visual acuity, pain-threshold tests
- Human Factors/Ergonomics — dashboard displays, alarms design
- Consumer/Marketing — smallest detectable change in price/packaging/taste (JND)
- Basic Sensory Research — mapping vision, hearing, taste, smell, touch

 Fechner founded psychophysics (1860) on Weber's work. Absolute threshold = detect 50% of time; JND = smallest detectable difference. Weber's Law (ratio) → Fechner's Law (log). SDT adds decision bias (d' , β).

Quick Revision Summary

- **Scope:** Psychology = science of behaviour & mental processes → **Theoretical** (pure research) vs **Applied** (real-world problem-solving); 15+ branches from experimental to community psychology.
- **Methods:** Only **Experimental** gives cause-effect (IV → DV, random assignment). Correlational = relationship, not causation. Case study = depth. Naturalistic obs. = real-world truth. Survey = breadth. Longitudinal (same people, over time) vs Cross-sectional (different ages, one time). Ethics: consent → withdrawal → debrief → confidentiality → protection.
- **Perception:** Sensation (data) → Perception (meaning). Bottom-up (Gibson, data → whole) vs Top-down (Gregory, concept → check data). Gestalt laws: proximity, similarity, closure, continuity, figure-ground. Depth = monocular (1 eye: perspective, size, overlap, texture, aerial, shadow) + binocular (2 eyes: disparity, convergence). Constancy = size/shape/colour/brightness stay stable. Illusion (real stimulus misread) ≠ Hallucination (no stimulus).
- **Psychophysics:** Fechner (1860) founded it on Weber's JND work. Absolute threshold = detected 50% of time. JND = smallest detectable difference. $\Delta I/I = k$
Weber's Law (proportion). $S = k \log I$ Fechner's Law (log growth). SDT → d' (sensitivity) + β (bias); 4 outcomes: Hit/Miss/False Alarm/Correct Rejection. 3 methods: Limits, Constant Stimuli, Adjustment.

— end of notes, all the best for exams!  —